# STPM 2023 PHYSICS 960/1

## Questions and Answers

*All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.*

## Section A [15 marks]

**1.** A string with a mass, $m$, and length, $l$, has a tension, $T$. Which quantity has the same dimension as $\sqrt{(\frac{Tl}{m})}$?

A Force B Velocity C Power D Energy

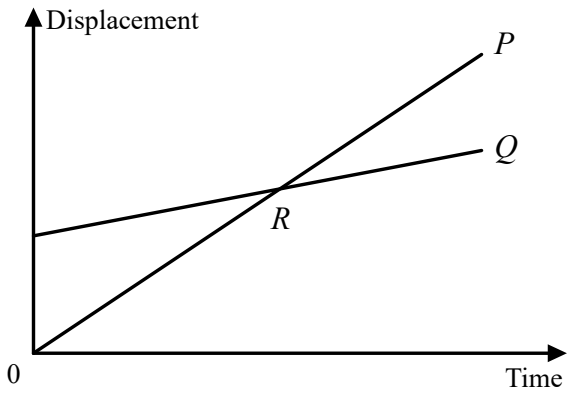
**Answer: B**

$$[\frac{Tl}{m}] = \frac{(\text{kg m s}^{-2})(\text{m})}{\text{kg}} = \text{m}^2\ \text{s}^{-2}$$

$$[\sqrt{(\frac{Tl}{m})}] = \text{m s}^{-1} \rightarrow \text{velocity}$$

**Explanation:** $m/l$ is the mass per unit length of the string, and $\sqrt{(\frac{T}{m/l})}$ is in fact the speed of a wave on a stretched string. Work with base units: the kilograms cancel and the square root of $\text{m}^2\ \text{s}^{-2}$ is $\text{m s}^{-1}$.

**2.** The motion of two cars, $P$ and $Q$, on a straight road are shown in the graph of displacement against time below.




Which statement is **not** true about the motion of both cars?

A $Q$ overtakes $P$ at $R$.
B $Q$ moves slower than $P$.
C $Q$ and $P$ start simultaneously.
D $Q$ and $P$ stop simultaneously.

**Answer: A**

The gradient of a displacement–time graph is the velocity. $P$ has the steeper line (faster) and starts from the origin; $Q$ starts ahead of $P$ but moves more slowly.

At $R$ the faster car $P$ catches up with and overtakes $Q$, so the statement "$Q$ overtakes $P$ at $R$" is not true.

**Explanation:** Where two displacement–time lines cross, the cars are at the same place at the same time. After $R$ the line of $P$ is above the line of $Q$, so $P$ is ahead: $P$ overtakes $Q$, not the other way round. Both lines begin at $t = 0$, so they start at the same time.

**3.** A particle hits a wall perpendicularly at speed $u$. Which statement is true if the collision is an elastic collision?

A The particle returns along its original path with speed $u$, while the total momentum and total kinetic energy are conserved.

B The particle returns along its original path with speed less than $u$, while the total momentum and total kinetic energy are conserved.

C The particle returns along its original path with speed $u$, while the total momentum is conserved but the total kinetic energy is not conserved.

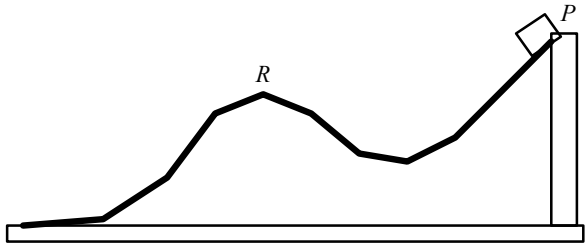
D The particle returns along its original path with speed more than $u$, while the total momentum is conserved but the total kinetic energy is not conserved.

**Answer: A**

In an elastic collision the total kinetic energy is conserved, so the particle rebounds along its original path with the same speed $u$; the total momentum (particle + wall) is also conserved.

**Explanation:** Momentum is conserved in every collision; kinetic energy is conserved only in an elastic collision. A speed less than $u$ would mean kinetic energy is lost (inelastic), and a speed more than $u$ is impossible without an energy source.

**4.** A trolley is released on a track at $P$ and moves through $R$ as shown in the diagram below.




If the heights at $P$ and $R$ are 1.2 m and 0.90 m respectively, what is the speed of the trolley at $R$?

A $2.4\ \text{m s}^{-1}$ B $4.2\ \text{m s}^{-1}$ C $4.9\ \text{m s}^{-1}$ D $6.4\ \text{m s}^{-1}$

**Answer: A**

$$\frac{1}{2}mv^2 = mg(h_P - h_R)$$

$$v = \sqrt{(2(9.81)(1.2 - 0.90))} = 2.4\ \text{m s}^{-1}$$

**Explanation:** Assuming no friction, the trolley's loss in potential energy between $P$ and $R$ becomes kinetic energy. Only the difference in height (0.30 m) matters, not the shape of the track. Using 1.2 m alone gives option C.

**5.** A small object is placed at different positions on a horizontal disc which rotates at constant angular velocity, $\omega$. When the distance of the object is $r$ from the centre of the disc, the linear speed of the object is $v$ and the centripetal acceleration is $a$. Which position of the object corresponds correctly to the linear speed and centripetal acceleration?

| | *Position* | *Linear speed* | *Centripetal acceleration* |
|---|---|---|---|
| **A** | $\frac{1}{2}r$ | $\frac{1}{2}v$ | $a$ |
| **B** | $\frac{1}{2}r$ | $2v$ | $\frac{1}{2}a$ |
| **C** | $2r$ | $\frac{1}{2}v$ | $2a$ |
| **D** | $2r$ | $2v$ | $2a$ |

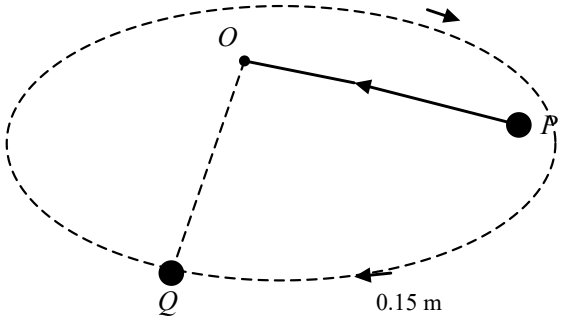
**Answer: D**

$v = \omega r$ and $a = \omega^2 r$; $\omega$ is the same at every point of the disc

Both $v$ and $a$ are directly proportional to $r$: at $2r$, the linear speed is $2v$ and the centripetal acceleration is $2a$.

**Explanation:** All points on a rotating rigid disc share the same angular velocity. With $\omega$ fixed, doubling the radius doubles both $v = \omega r$ and $a = \omega^2 r$; halving the radius halves both. Using $a = \frac{v^2}{r}$ alone is misleading because $v$ also changes.

**6.** An object of mass 200 g is tied up to one end of a string and the object is whirled round in a circle on a smooth horizontal surface as shown in the diagram below.




The tension in the string is 2.4 N and the velocity of the object is 1.2 m s$^{-1}$. If the circular path from $P$ to $Q$ is 0.15 m, what is the angular displacement of the object?

A 0.018 rad B 0.80 rad C 1.3 rad D 2.1 rad

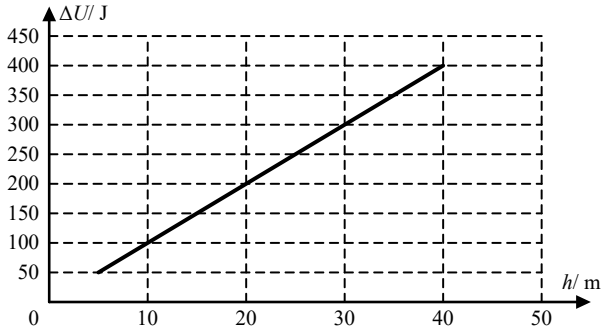
**Answer: C**

$$T = \frac{mv^2}{r} \Rightarrow r = \frac{mv^2}{T} = \frac{(0.200)(1.2)^2}{2.4} = 0.12 \text{ m}$$

$$\theta = \frac{s}{r} = \frac{0.15}{0.12} = 1.3 \text{ rad}$$

**Explanation:** On a smooth horizontal surface the tension is the only horizontal force, so it provides the centripetal force; this gives the radius. The angle in radians is the arc length divided by the radius. Convert 200 g to 0.200 kg.

**7.** The variation of the change in gravitational potential energy, $\Delta U$, with the height, $h$, above the surface of a planet of a body of mass 2.0 kg is shown in the graph below.




What is the acceleration due to gravity at the surface of the planet?

A $5.0\ \text{m s}^{-2}$ B $10\ \text{m s}^{-2}$ C $18\ \text{m s}^{-2}$ D $20\ \text{m s}^{-2}$

**Answer: A**

$$\Delta U = mgh \Longrightarrow \text{gradient of the } \Delta U\text{–}h \text{ graph} = mg$$

$$\text{Gradient} = \frac{400 - 100}{40 - 10} = 10\ \text{J m}^{-1} \Longrightarrow g = \frac{10}{2.0} = 5.0\ \text{m s}^{-2}$$

**Explanation:** Near the surface of the planet $g$ is constant, so the change in potential energy is proportional to the height and the graph is a straight line. Its gradient is the weight of the body; divide by the mass to get $g$. Option B (10) is the gradient, not $g$.

**8.** A satellite moves around the Earth in a circular orbit. Which is correct about the satellite?

A The speed varies
B The kinetic energy varies
C The linear momentum varies
D The angular velocity varies

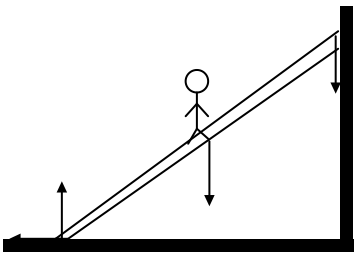
**Answer: C**

In a circular orbit the speed, kinetic energy and angular velocity are constant, but the direction of motion changes continuously, so the linear momentum (a vector) varies.

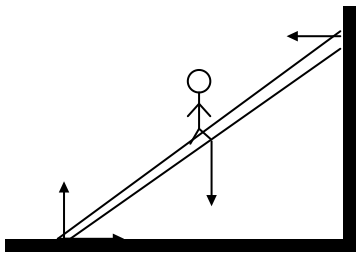
**Explanation:** Momentum $p = mv$ is a vector with the same direction as the velocity. A satellite in a circular orbit keeps a constant speed, so its kinetic energy and angular velocity do not change, but its direction of motion keeps turning, so its momentum is always changing.

**9.** A man stands on a light ladder that is placed on a rough floor and leaned against a rough wall. Which diagram shows the correct direction of the forces acting on the ladder?

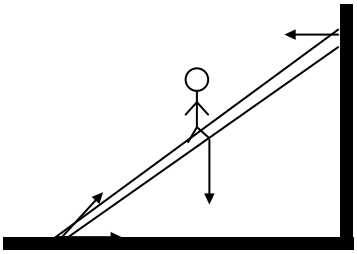
A



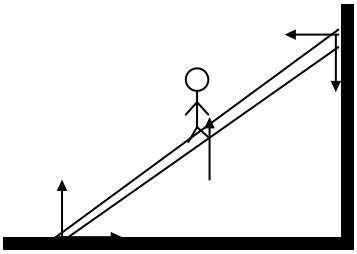
B



C



D

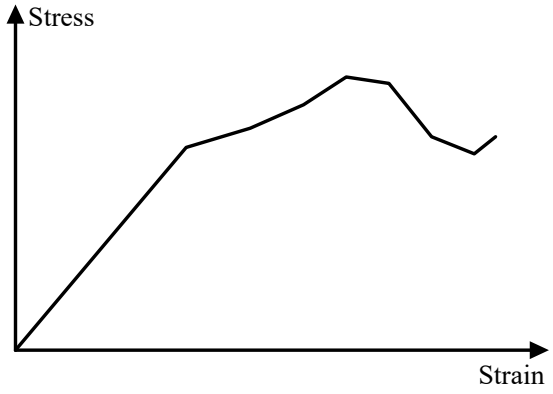


**Answer: B**

Forces on the ladder: weight of the man (down); normal reaction of the floor (up); friction at the floor towards the wall; normal reaction of the wall, perpendicular to the wall (away from it). Diagram B shows these directions.

**Explanation:** A normal reaction is always perpendicular to the surface: vertical at the floor and horizontal at the wall. The foot of the ladder tends to slide away from the wall, so friction at the floor acts towards the wall. The wall pushes the ladder away from the wall; it cannot pull it.

**10.** The variation of stress with strain of a wire is shown in the graph below.




Which is represented by the area under the graph?

A Force constant of the wire
B Young modulus of the wire
C Elastic potential energy in the wire
D Strain energy per unit volume of the wire

**Answer: D**

$$\text{Area} = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}\left(\frac{F}{A}\right)\left(\frac{e}{l}\right) = \frac{\frac{1}{2}Fe}{Al} = \frac{\text{strain energy}}{\text{volume}}$$

**Explanation:** Stress × strain = $\frac{Fe}{Al}$, and $Al$ is the volume of the wire, so the area under a stress–strain graph is energy per unit volume. The gradient of the linear part (not the area) gives the Young's modulus, and the area under a force–extension graph gives the total elastic potential energy.

**11.** The average translational kinetic energy of an oxygen molecule is $3.15 \times 10^{-21}$ J. What is the temperature of the oxygen gas?

A 65.0 K B 76.0 K C 91.3 K D 152 K

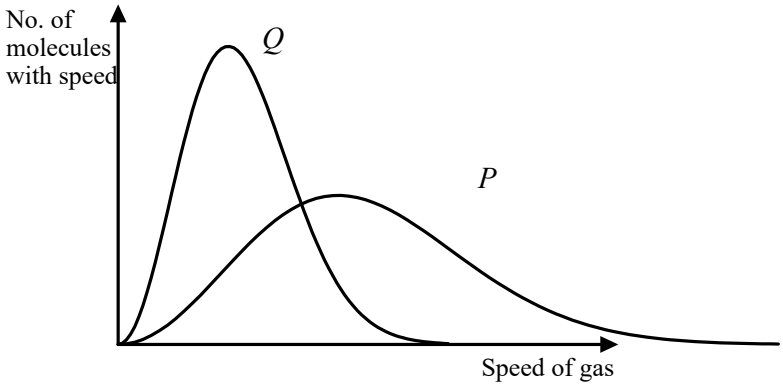
**Answer: D**

$$\frac{3}{2}kT = 3.15 \times 10^{-21} \text{ J}$$

$$T = \frac{2(3.15 \times 10^{-21})}{3(1.38 \times 10^{-23})} = 152 \text{ K}$$

**Explanation:** The average translational kinetic energy of any gas molecule is $\frac{3}{2}kT$, whatever the gas. Oxygen is diatomic, but the question gives only the translational part, so do not use $\frac{5}{2}kT$ (which would give option C).

**12.** The curves of molecular speed distribution for a gas is shown in the graph below.




Which action would cause the change of the curve from *P* to *Q*?

A The gas is heated in a sealed container.
B The gas is cooled in a sealed container.
C The gas is heated and some of the molecules leave the container.
D The gas is cooled and some of the molecules leave the container.

**Answer: B**

Curve *Q* has its peak at a lower speed and is narrower and taller than *P*, while the area under each curve (the number of molecules) is the same: the gas has been cooled without losing molecules.

**Explanation:** When a gas is cooled the most probable speed decreases, so the peak moves to the left; fewer molecules have high speeds, so the curve becomes narrower and taller. Because the container is sealed, the total number of molecules, given by the area under the curve, does not change.

**13.** A closed cylinder which contains one mole of gas molecules is heated from 27.0 °C to 177 °C at constant pressure. What is the work done by the gas to increase the volume of the gas?

A 224 J B 1250 J C 1470 J D 1700 J

**Answer: B**

$$W = p\Delta V = nR\Delta T$$
$$W = (1)(8.31)(177 - 27.0) = 1250 \text{ J}$$

**Explanation:** At constant pressure $pV = nRT$ gives $p\Delta V = nR\Delta T$, so the work done by the gas depends only on the change in temperature. A temperature change of 150 °C is the same as 150 K.

**14.** A monoatomic ideal gas is initially at a pressure of $1.50 \times 10^4$ Pa and a volume of 0.0900 $m^3$. The gas is compressed adiabatically to 0.0400 $m^3$. What is the final pressure of the gas?

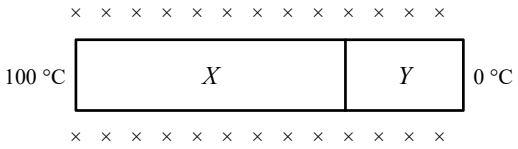
A $3.87 \times 10^3$ Pa B $4.82 \times 10^3$ Pa C $4.67 \times 10^4$ Pa D $5.81 \times 10^4$ Pa

**Answer: D**

$$p_1V_1^{\gamma} = p_2V_2^{\gamma}, \text{ with } \gamma = \frac{5}{3} \text{ for a monoatomic gas}$$
$$p_2 = (1.50 \times 10^4)\left(\frac{0.0900}{0.0400}\right)^{5/3} = 5.81 \times 10^4 \text{ Pa}$$

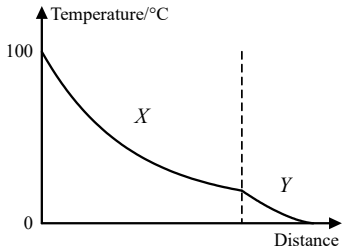
**Explanation:** For an adiabatic process $pV^{\gamma}$ is constant. A monoatomic gas has 3 degrees of freedom, so $\gamma = \frac{C_p}{C_V} = \frac{5}{3}$. The gas is compressed, so its pressure must increase: options A and B can be ruled out at once.

**15.** Two well-insulated metal rods, *X* and *Y*, with an equal cross-sectional area joined end-to-end as shown in the diagram below.

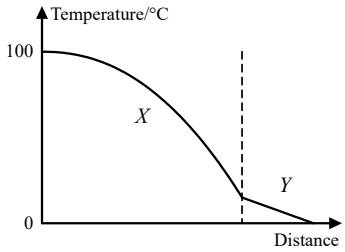



Both ends of *X* and *Y* are maintained at 100 °C and 0 °C respectively. At steady state, which graph represents the variation of temperature with distance along the rods?

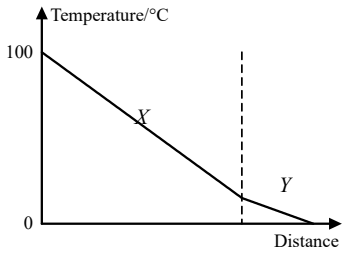
A



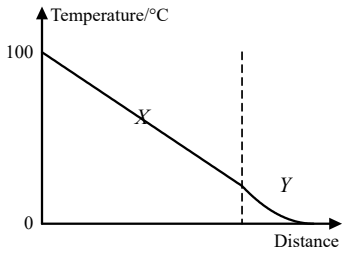

B



C




D



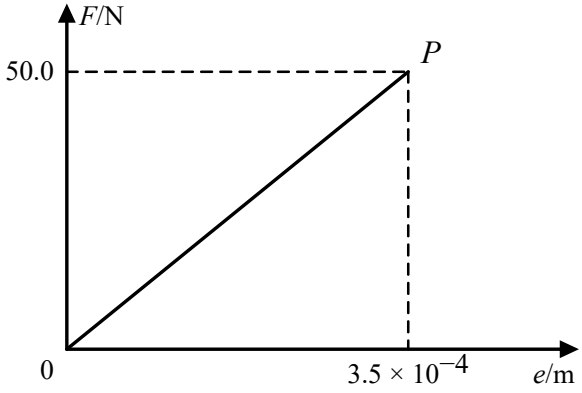

**Answer: C**

In the steady state the same rate of heat flow passes through $X$ and $Y$ (well insulated, joined in series), so within each rod the temperature falls **linearly** with distance. The two straight lines have different gradients and meet at the junction.

**Explanation:** From $\frac{Q}{t} = kA\frac{\Delta\theta}{\Delta x}$, a constant rate of heat flow through a uniform, insulated rod needs a constant temperature gradient, so the graph in each rod is a straight line. Curved graphs (A, B and part of D) would mean heat is lost from the sides of the rods.

## Section B [15 marks]

**16.** The variation of tensile force, $F$, with extension, $e$, of a glass fibre until the fibre breaks at $P$ is shown in the graph below.




The length and the cross-sectional area of the glass fibre are 0.28 m and $6.4 \times 10^{-7}$ m$^2$ respectively.

(a) State whether the glass fibre is brittle or ductile. Give a reason for your answer. [2]

**Answer**

| | |
|---|---|
| Brittle | 1 |
| No plastic deformation before the fibre breaks (the graph is a straight line up to the breaking point) | 1 |
| | **2** |

**Explanation:** A brittle material breaks soon after its elastic limit without any plastic deformation, so its force–extension graph is a straight line that ends suddenly. A ductile material would show a long curved plastic region before breaking.

(b) Determine the maximum tensile stress. [2]

**Answer**

Tensile stress, $\sigma = \frac{F}{A} = \frac{50.0}{6.4 \times 10^{-7}}$ 1

$\sigma = 7.81 \times 10^{7}\ \text{N m}^{-2}$ 1

*[Accept $7.8 \times 10^{7}$ or $7.813 \times 10^{7}$; unit Pa accepted]*

**2**

**Explanation:** The maximum stress occurs at the breaking point *P*, where the force is 50.0 N. Stress is force per unit cross-sectional area.

(c) Determine the Young's modulus for the glass fibre. [2]

**Answer**

$E = \frac{\sigma}{e/L} = \frac{\text{stress}}{\text{strain}}$ 1

$E = \frac{7.81 \times 10^{7}}{\frac{3.5 \times 10^{-4}}{0.28}} = 6.25 \times 10^{10}\ \text{N m}^{-2}$ 1

*[Accept $6.2 \times 10^{10}$ or $6.248 \times 10^{10}$; or accordingly from (b)]*

**2**

**Explanation:** The graph is a straight line all the way, so the stress and strain at the breaking point can be used. Strain = extension ÷ original length (0.28 m).

(d) Calculate the energy released once the glass fibre breaks. [2]

**Answer**

Energy = $\frac{1}{2}Fe$ (area under the graph) 1

$= \frac{1}{2}(50.0)(3.5 \times 10^{-4}) = 8.75 \times 10^{-3}\ \text{J}$ 1

*[Accept $8.8 \times 10^{-3}$ J; alternative: $U = \frac{1}{2}ke^2$ with $k = \frac{F}{e}$]*

**2**

*Total for question* **8**

**Explanation:** The energy stored in the fibre is the area under the force–extension graph, a triangle here. Because the fibre is brittle, all of this elastic energy is released suddenly when it breaks.

**17.** (a) The ozone layer in the Earth atmosphere consists of ozone gas, $O_3$, that has the density of 0.20 kg $m^{-3}$ and temperature −56 °C. Assume that the gas behaves as an ideal gas. Calculate the pressure of the ozone layer.
[Molar mass of ozone is 48 g $mol^{-1}$.] [4]

**Answer**

$pV = nRT$ 1

$n = \frac{m}{M} \Longrightarrow p = \frac{m}{MV}RT$ 1

$\rho = \frac{m}{V} \Longrightarrow p = \frac{\rho RT}{M}$ 1

$p = \frac{(0.20)(8.31)(217)}{48 \times 10^{-3}} = 7.51 \times 10^3$ Pa 1

*[$T = -56 + 273 = 217$ K. Alternative: $p = \frac{1}{3}\rho\langle c^2\rangle$ with $\langle c^2\rangle = \frac{3RT}{M}$]*

**4**

**Explanation:** Rewrite the ideal gas equation in terms of density: $p = \frac{\rho RT}{M}$. Convert the temperature to kelvin and the molar mass to kg $mol^{-1}$ ($48 \times 10^{-3}$).

(b) State **three** ways to reduce global warming. [3]

**Answer**

• Reduce the use of fossil fuels / reduce the production of $CO_2$ (e.g. car pooling) 1

• Avoid open burning 1

• Plant more trees to absorb carbon dioxide 1

• Use more renewable or clean energy (solar, wind, hydro, tidal, nuclear, biodiesel) 1

*[Any three – maximum 3 marks]*

**3**

*Total for question* **7**

**Explanation:** Global warming is caused by an increase in greenhouse gases, mainly carbon dioxide and methane. Ways to reduce it either produce less of these gases (less burning of fossil fuels, cleaner energy) or remove carbon dioxide from the air (planting trees).

## Section C [30 marks]

**18.** (a) An object is released from rest in mid air. Explain the effect of air resistance on the motion of the object in the air. [7]

**Answer**

| | |
|---|---|
| Resultant force on the object: $F = mg - R$ ($R$ = air resistance) | 1 |
| $F = ma$, so $mg - R = ma$ | 1 |
| When the object is released, its velocity increases (it accelerates) | 1 |
| The air resistance on the object increases as the velocity increases | 1 |
| So the resultant force and the acceleration of the object decrease | 1 |
| When $R = mg$, the resultant force is zero and $a = 0$ | 1 |
| The object then moves with a constant velocity, the terminal velocity | 1 |
| | **7** |

**Explanation:** Air resistance increases with speed. At first only the weight acts, so the object accelerates at $g$; as it speeds up, the air resistance grows and the acceleration gets smaller, until the air resistance balances the weight. The velocity–time graph is a curve that levels off at the terminal velocity.

(b) A drone on a live sports event is flying horizontally at a speed of 30.0 m $s^{-1}$ and accidentally drops a camera at the height of 120 m. Neglect the air resistance.

(i) Determine the time taken for the camera to reach the ground. [3]

**Answer**

| | |
|---|---|
| $s = ut + \frac{1}{2}at^2$ | 1 |
| Initial vertical velocity $u = 0$ | 1 |
| $t = \sqrt{(\frac{2s}{a})} = \sqrt{(\frac{2(120)}{9.81})} = 4.95$ s | 1 |
| | **3** |

**Explanation:** The camera leaves the drone with the drone's horizontal velocity, so its initial vertical velocity is zero. The time to fall depends only on the height, not on the horizontal speed.

(ii) Determine the horizontal distance of the camera before the camera lands on the ground. [2]

**Answer**

| | |
|---|---|
| $s = ut = (30.0)(4.95)$ | 1 |
| $s = 148.5$ m | 1 |
| *[Accept 149 m or 148.4 m]* | |
| | **2** |

**Explanation:** With no air resistance there is no horizontal force, so the horizontal velocity stays at 30.0 m $s^{-1}$ for the whole fall.

(iii) Calculate the magnitude of the velocity of the camera just before the camera lands on the ground. [3]

**Answer**

$v_x = 30.0$ m $s^{-1}$; $v_y = 0 + (9.81)(4.95) = 48.6$ m $s^{-1}$ 1

$v = \sqrt{(v_x^2 + v_y^2)} = \sqrt{(30.0^2 + 48.6^2)}$ 1

$v = 57.1$ m $s^{-1}$ 1

*[Alternative: conservation of energy,* $\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mgh$*]*

**3**

*Total for question* **15**

**Explanation:** Find the horizontal and vertical components of the velocity separately and combine them with Pythagoras' theorem. The question asks only for the magnitude, so the direction is not needed.

**19.** (a) Define Newton's law of universal gravitation. [1]

**Answer**

The gravitational (attractive) force between two masses is directly proportional to the product of the two masses and inversely proportional to the square of the distance between them; $F = \frac{GMm}{r^2}$. 1

**1**

**Explanation:** State both proportionalities and that the force is attractive. If you write the equation, define every symbol: $G$ is the gravitational constant, $M$ and $m$ are the masses and $r$ is the distance between their centres.

(b) Explain the change of gravitational acceleration when a meteoroid at a distance $3R$ from the surface of the Earth falls towards the surface of the Earth, where $R$ is the radius of the Earth. [6]

**Answer**

$F = mg$ 1

$\frac{GMm}{r^2} = mg \Longrightarrow g = \frac{GM}{r^2}$ 1

At distance $3R$ from the surface: $r = R + 3R = 4R$ 1

$g' = \frac{GM}{(4R)^2} = \frac{1}{16}g = 0.61 \text{ m s}^{-2}$ 1

$\Delta g = g - g'$ 1

$\Delta g = \frac{15}{16}g = 9.20 \text{ m s}^{-2}$: the gravitational acceleration increases as the meteoroid falls towards the surface 1

**6**

**Explanation:** The distance in $g = \frac{GM}{r^2}$ is measured from the centre of the Earth, so a point $3R$ above the surface is $4R$ from the centre. Because $g \propto \frac{1}{r^2}$, the acceleration there is only $\frac{1}{16}$ of its surface value, and it increases steadily as the meteoroid approaches the Earth.

(c) Juno satellite is launched from the surface of the Earth to Jupiter and begins orbiting Jupiter at the radius of $7.05 \times 10^4$ km. Given that the mass of Jupiter is $1.90 \times 10^{27}$ kg. Determine

(i) the escape speed of the satellite from the surface of the Earth [2]

**Answer**

$v = \sqrt{\left(\frac{2GM}{R}\right)} = \sqrt{\left(\frac{2(6.67 \times 10^{-11})(5.97 \times 10^{24})}{6.38 \times 10^6}\right)}$ 1

$v = 1.12 \times 10^4 \text{ m s}^{-1}$ 1

*[Using $v = \sqrt{(2gR)}$ gives $1.119 \times 10^4$ m $s^{-1}$, also accepted]*

**2**

**Explanation:** The escape speed is the minimum launch speed for which the kinetic energy equals the energy needed to reach infinity: $\frac{1}{2}mv^2 = \frac{GMm}{R}$. It does not depend on the mass of the satellite.

(ii) the distance travelled by the satellite if the satellite takes 1825 days to reach Jupiter [2]

**Answer**

$s = vt$ 1

$s = (1.12 \times 10^4)(1825 \times 24 \times 3600) = 1.77 \times 10^{12}$ m 1

*[ECF from (i)]*

**2**

**Explanation:** Assume the satellite travels at the escape speed found in (i). Convert 1825 days into seconds before multiplying.

(iii) the gravitational potential of satellite at its orbit [2]

**Answer**

$V = -\frac{GM_J}{r} = -\frac{(6.67 \times 10^{-11})(1.90 \times 10^{27})}{7.05 \times 10^{7}}$ 1

$V = -1.80 \times 10^{9}\ \text{J kg}^{-1}$ 1

*[The negative sign is compulsory]*

**2**

**Explanation:** Gravitational potential is negative because the zero of potential is at infinity and gravity is attractive. Convert $7.05 \times 10^{4}$ km into $7.05 \times 10^{7}$ m.

(iv) the gravitational field strength of the satellite due to Jupiter [2]

**Answer**

$g = \frac{GM_J}{r^2} = \frac{(6.67 \times 10^{-11})(1.90 \times 10^{27})}{(7.05 \times 10^{7})^2}$ 1

$g = 25.5\ \text{m s}^{-2}\ (\text{N kg}^{-1})$ 1

*[Alternative:* $g = \frac{|V|}{r} = \frac{1.798 \times 10^{9}}{7.05 \times 10^{7}} = 25.5\ m\ s^{-2}$*]*

**2**

*Total for question* **15**

**Explanation:** The gravitational field strength is the force per unit mass, $\frac{GM}{r^2}$. Since $V = -\frac{GM}{r}$, the magnitude of the field strength is also $\frac{|V|}{r}$, a quick check on (iii).

**20.** (a)

(i) State the law of equipartition of energy. [2]

**Answer**

The energy of a molecule is shared equally between all its degrees of freedom 1

Each degree of freedom has an average energy of $\frac{1}{2}kT$ 1

**2**

**Explanation:** A degree of freedom is an independent way in which a molecule can store energy (moving in $x$, $y$, $z$, rotating about an axis, …). Equipartition says each one gets the same average share, $\frac{1}{2}kT$.

(ii) Derive an expression for the internal energy of $n$ mol of ideal gas with degrees of freedom, $f$, and temperature, $T$. [3]

**Answer**

Kinetic energy of one molecule $= \frac{f}{2}kT$ — 1

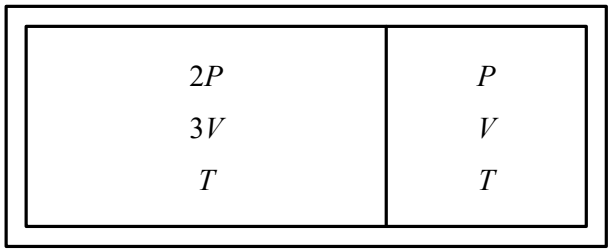
$U = \frac{f}{2}kT \times N$, with $N = nN_A$ — 1

$U = \frac{f}{2}kTnN_A = \frac{f}{2}nRT$ (since $R = N_Ak$) — 1

**3**

**Explanation:** The internal energy of an ideal gas is the total kinetic energy of all its molecules (there are no intermolecular forces, so no potential energy). Multiply the energy of one molecule by the number of molecules.

(b) A container with insulating wall is divided into two compartments by a partition as shown in the diagram below.




The partition is breakable. The container is filled with oxygen. The pressure, volume and temperature in the right compartment are $P$, $V$ and $T$ while in the left compartment are $2P$, $3V$ and $T$.

(i) Determine the ratio of the number of molecules in the left compartment to that in the right compartment. [2]

**Answer**

$PV = nRT \Longrightarrow n_R = \frac{PV}{RT}$, $n_L = \frac{(2P)(3V)}{RT}$ — 1

$\frac{n_L}{n_R} = 6$, i.e. 6 : 1 — 1

**2**

**Explanation:** Both compartments are at the same temperature, so the number of molecules is proportional to $pV$. The ratio of the numbers of molecules equals the ratio of the numbers of moles.

(ii) Calculate the pressure of the gas in the container after the partition breaks. [2]

**Answer**

Total volume = $4V$; total number of moles = $n_L + n_R$: $(2P)(3V) + PV = P_f(4V)$ — 1

$P_f = \frac{7}{4}P = 1.75P$ — 1

**2**

**Explanation:** The walls are insulating and no work is done, so the temperature stays at $T$. The total number of moles is conserved, so the sum of $pV$ before equals $pV$ after the partition breaks.

(iii) Calculate the internal energy of the gas in the container if the initial values of $P$, $V$ and $T$ in the right compartment are $1.014 \times 10^5$ Pa, $5.0 \times 10^{-4}$ m$^3$ and 303 K respectively. [2]

**Answer**

$U = \frac{f}{2}PV$, with $f = 5$ for oxygen (diatomic) — 1

$U = \frac{5}{2} \times \frac{7}{4}(1.014 \times 10^5) \times 4 \times 5.0 \times 10^{-4} = 887$ J — 1

*[Accept 890 J or $8.9 \times 10^2$ J; alternative: $U = \frac{5}{2}nRT$]*

**2**

**Explanation:** Since $nRT = PV$, the internal energy $\frac{f}{2}nRT$ can be written as $\frac{f}{2}PV$. Use the final pressure $1.75P$ and the total volume $4V$, and $f = 5$ because oxygen is diatomic.

(iv) If the gas behave as a real gas, discuss whether the internal energy of the gas is greater or smaller than the calculated value in (b)(iii). [4]

**Answer**

The internal energy of the real gas is smaller than that of the ideal gas — 1

because the molecules of a real gas have kinetic energy and (negative) potential energy — 1

due to the intermolecular (attractive and repulsive) forces — 1

but the molecules of an ideal gas have only kinetic energy — 1

**4**

*Total for question* **15**

**Explanation:** In a real gas the attractive forces between molecules give them a negative potential energy, which lowers the total internal energy compared with an ideal gas at the same temperature, where the internal energy is kinetic energy only.